

15-12-2024
Sunday

used for
pattern
matching

Finite Automata: $\begin{cases} \rightarrow \text{DFA} \\ \rightarrow \text{NFA} \end{cases}$

Tuple: $\{Q, q, \Sigma, F, \delta\}$

Q : States: $\{q_1, q_2\}$

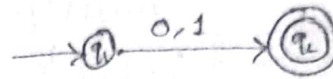
q : Initial state: q_1

Σ : input symbols: $\{0, 1\}$

F : set of final states: $\{q_2\}$

δ : Transition function:

States	Symbol	Destination
q_1	0	q_2
q_1	1	q_2



	0	1
q_1	q_2	q_2
q_2	$\emptyset$	$\emptyset$

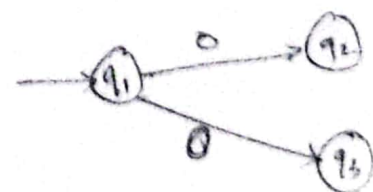
Difference of DFA and NFA:

DFA

- No empty transition present
- one transition for one input symbol

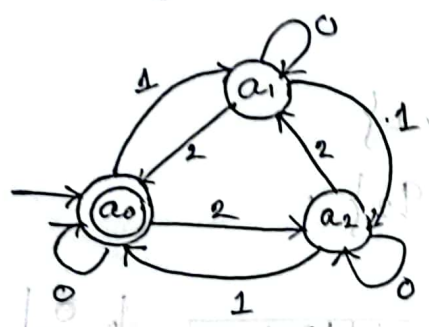
NFA

- empty transition is present
- multi transition for same input symbol

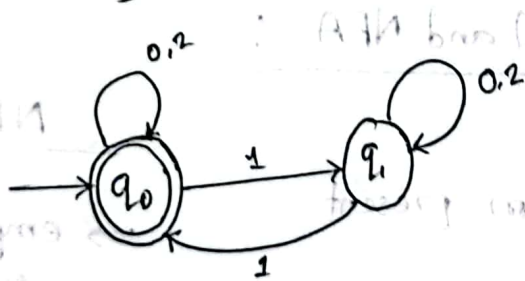


DFA Design:

Language: $A_1 = \{w: \text{the sum of all the symbols in } w \text{ is multiple of } 3\}$
 Alphabet: $\Sigma = \{0, 1, 2\}$



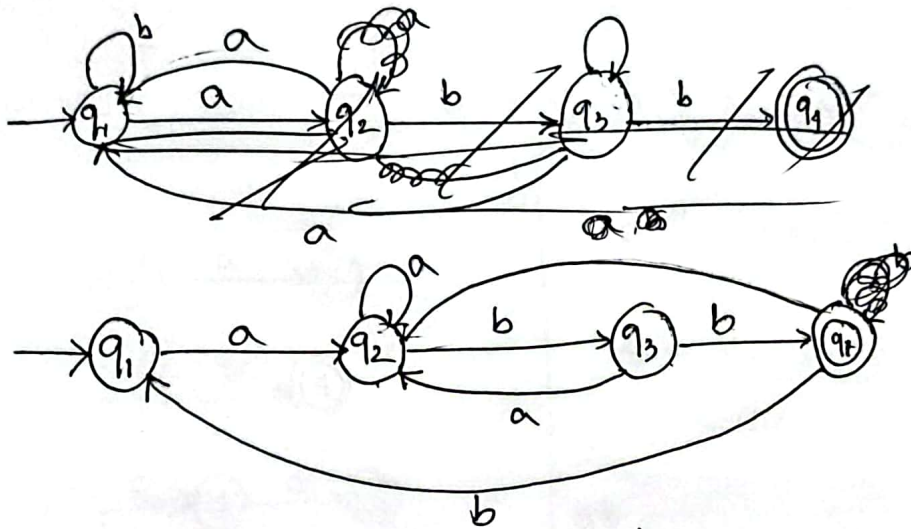
Language: $A_2 = \{w: \text{the sum of all the symbols in } w \text{ is an even number}\}$
 Alphabet: $\Sigma = \{0, 1, 2\}$



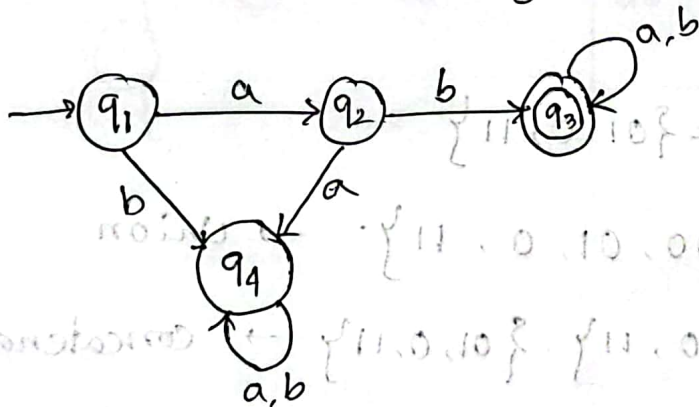
① Language, $A_1 = \{w: \text{strings ending with 'abb'}\}$
 Alphabet, $\Sigma = \{a, b\}$

② Language, $A_2 = \{w: \text{strings starting with 'ab'}\}$
 Alphabet, $\Sigma = \{a, b\}$

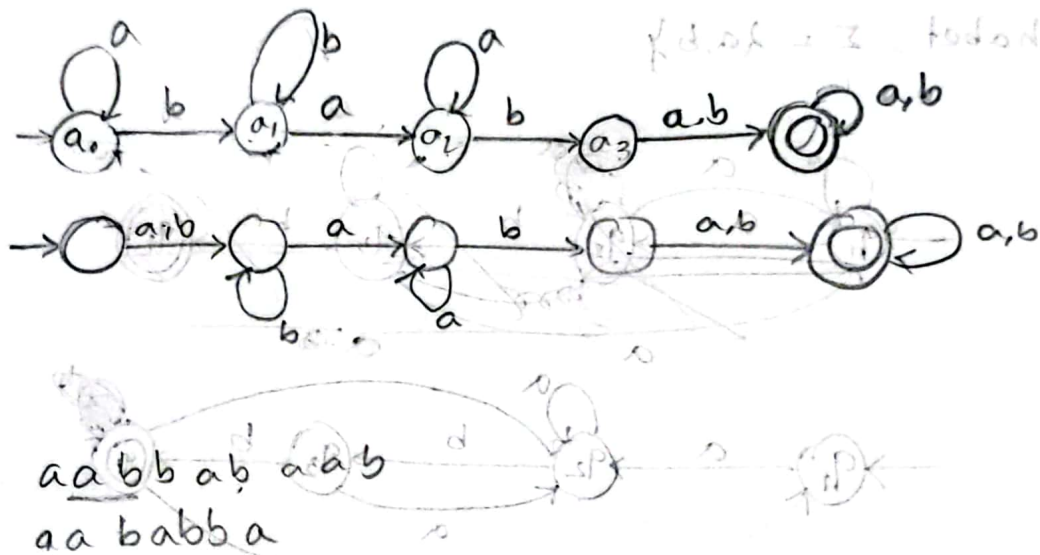
①



②



③ Language: $A_3 = \{ \text{accepted string with 'ab' in the middle} \}$
(sub string)
Symbol, $\Sigma = \{a, b\}$



Regular Expression

$$L = \{00, 11\} \quad M = \{01, 0, 11\}$$

$$L \cup M / L \mid M = \{00, 01, 0, 11\} \rightarrow \text{Union}$$

$$L \cdot M = \{00, 11\} \cdot \{01, 0, 11\} \rightarrow \text{Concatenation}$$

$$= \{0001, 000, 0011, 1101, 110, 1111\}$$

$$L^0 = \{\epsilon\}$$

$$L^1 = \{00, 11\}$$

$$L^2 = L \cdot L$$

$$= \{00, 11\} \cdot \{00, 11\}$$

$$= \{0000, 0011, 1100, 1111\}$$

L^* $\rightarrow$ ϵ and any combination of L

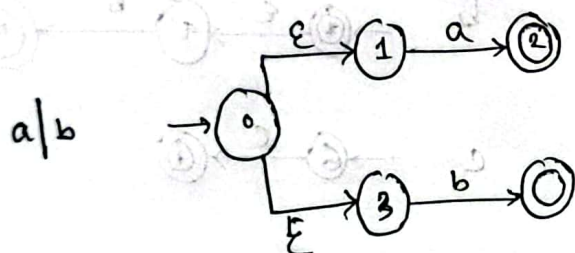
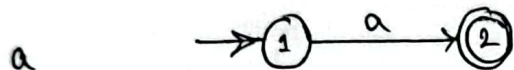
$$L^* = \{\epsilon, L^0, L^1, L^2, \dots, L^N\}$$

$$= \{\epsilon, 00, 11, 0000, 0011, \dots\}$$

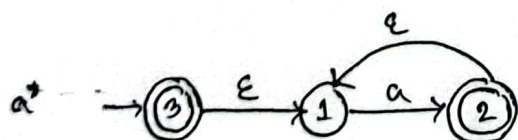
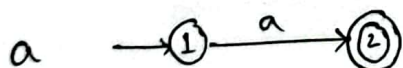
$L^+ = \{L^1, L^2, L^3, \dots, L^N\} \rightarrow$ accepts anything except ϵ/L^0 ^{combination of L}

Regular Expression to NFA :

a/b



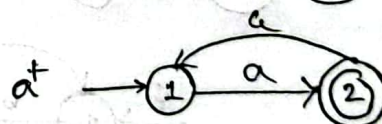
a^*



$a.b$

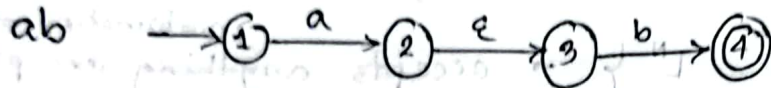


a^+

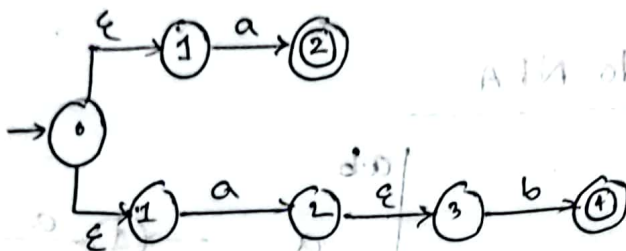


~~* > + > .~~
 $\boxed{+, * > . > |}$ → precedence

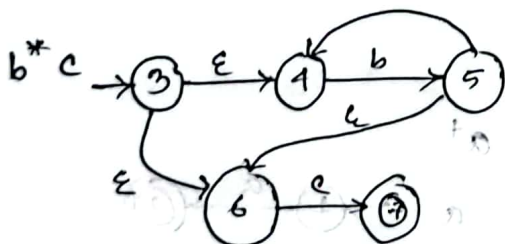
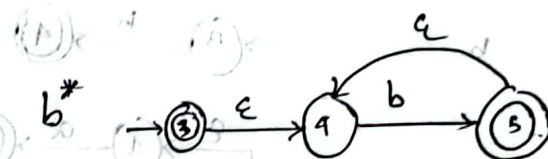
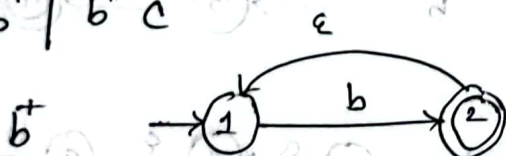
1) a/ab



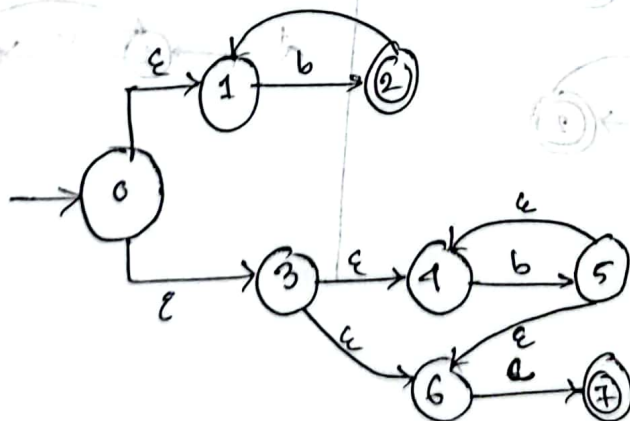
a/ab



2) $b^+ / b^* c$

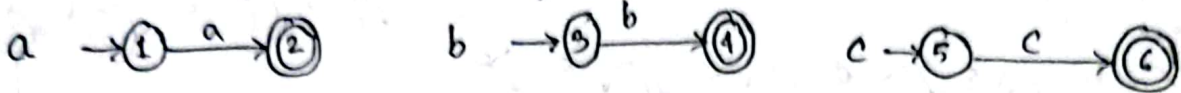


$b^+ / b^* c$

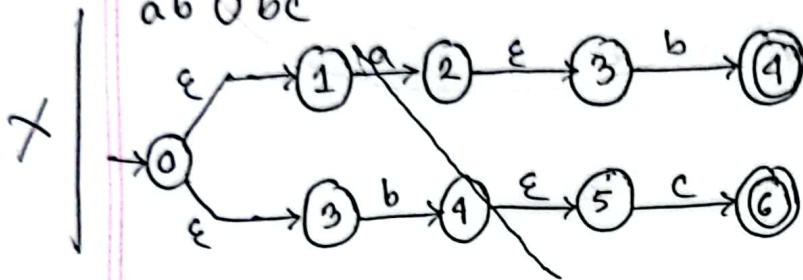


22-12-2024
Sunday

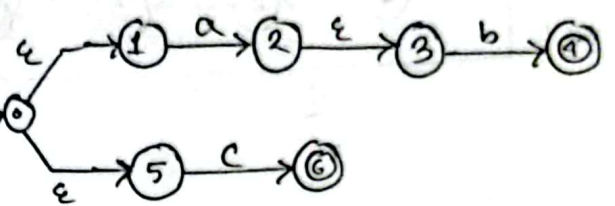
$(ab \cup bc (ab \cup c)^*)^*$



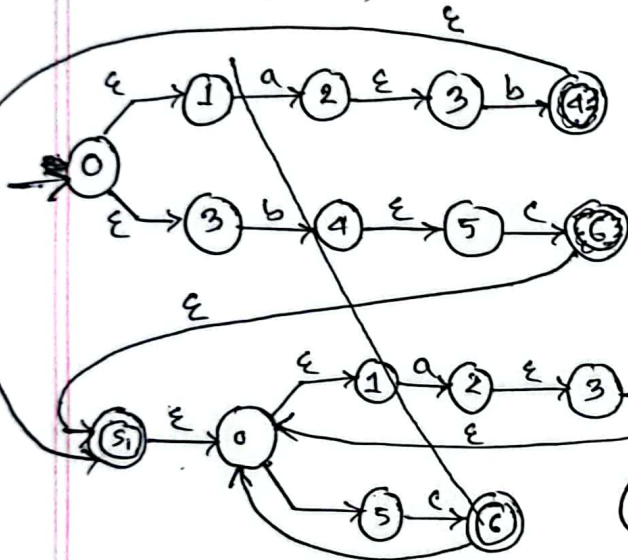
$ab \cup bc$



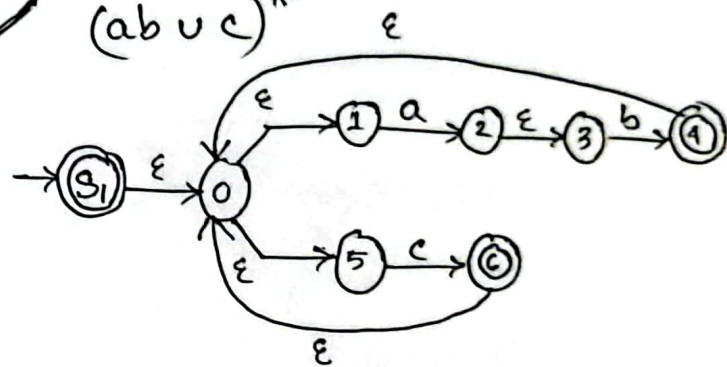
$ab \cup c$



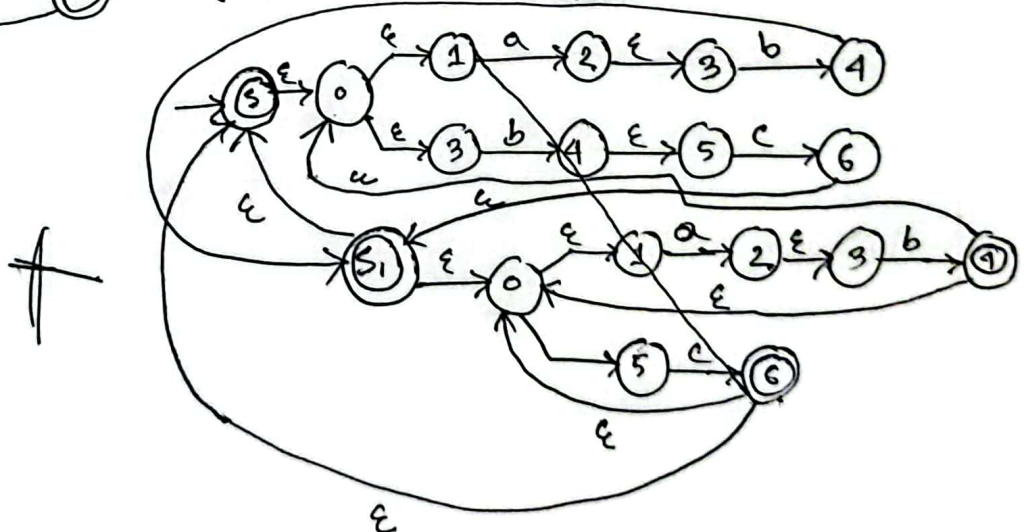
$ab \cup bc (ab \cup c)^*$



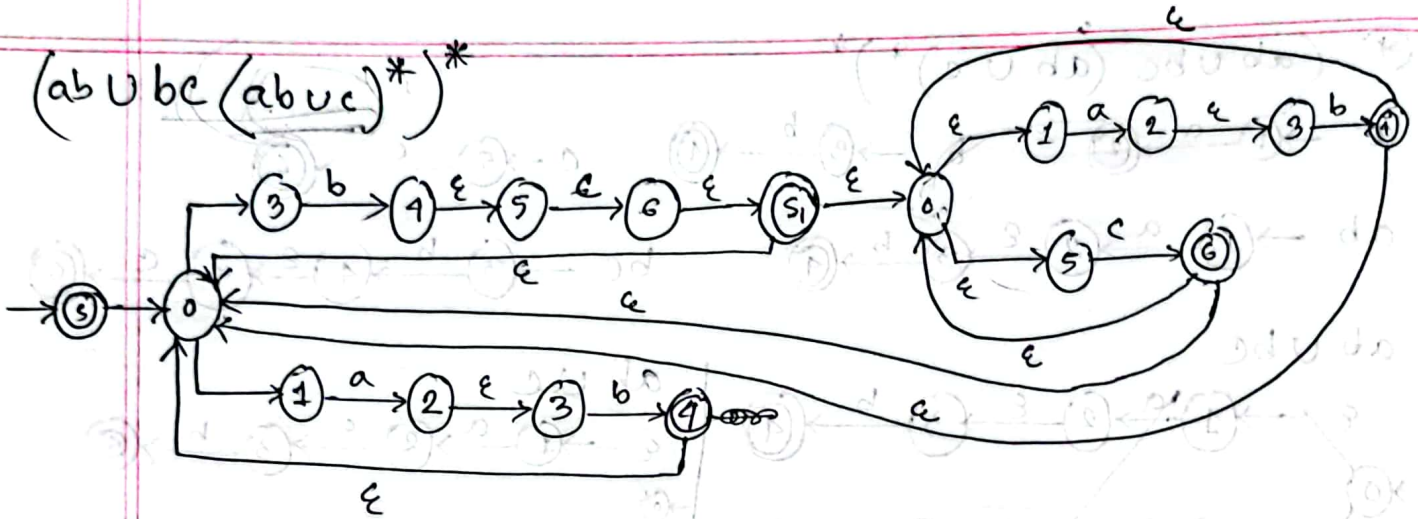
$(ab \cup c)^*$



$(ab \cup bc (ab \cup c)^*)^*$



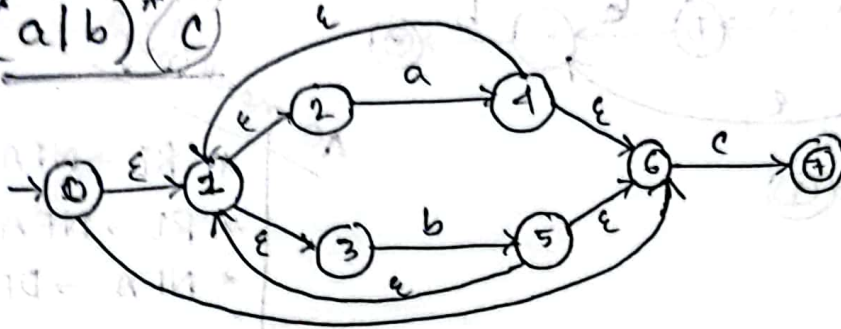
$(ab \cup bc (abuc)^*)^*$



RE $\rightarrow$ NFA $\rightarrow$ DFA

Subset construction Method

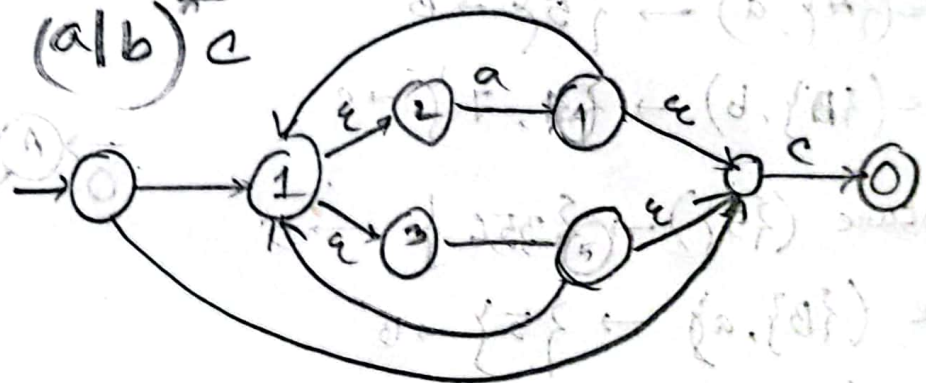
(1) $(a|b)^*c$



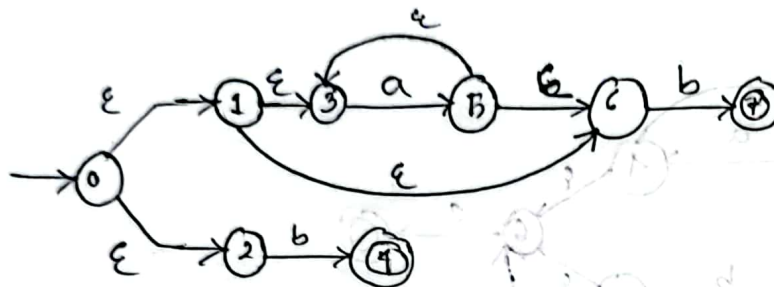
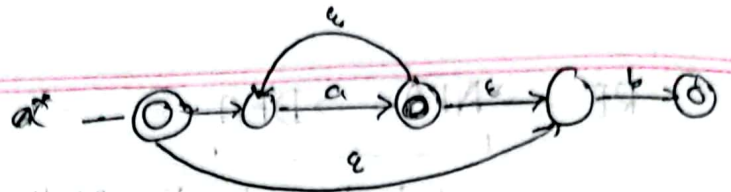
DFA

$\epsilon - c$

$(a|b)^*c$



2) $a^* + a^*b|b$



40 marks (min)

* RE $\rightarrow$ NFA $\rightarrow$ DFA
 * RE $\rightarrow$ NFA
 * NFA $\rightarrow$ DFA

DFA

ϵ -closure ($\{0\}$) $\rightarrow \{0, 1, 2, 3, 6\} \rightarrow A$

Move ($\{A\}, a$) $\rightarrow \{5\} \rightarrow B$

Move ($\{B\}, b$) $\rightarrow \{4, 7\} \rightarrow C$

ϵ -closure ($\{5\}$) $\rightarrow \{3, 5, 6\} \rightarrow B$

Move ($\{B\}, a$) $\rightarrow \{5\} \rightarrow B$

Move ($\{B\}, b$) $\rightarrow \{7\} \rightarrow D$

ϵ -closure ($\{4, 7\}$) $\rightarrow \{4, 7\} \rightarrow C$

Move ($\{C\}, a$) $\rightarrow \emptyset$

Move ($\{C\}, b$) $\rightarrow \emptyset$

ϵ -closure ($\{7\}$) $\rightarrow \{7\} \rightarrow D$

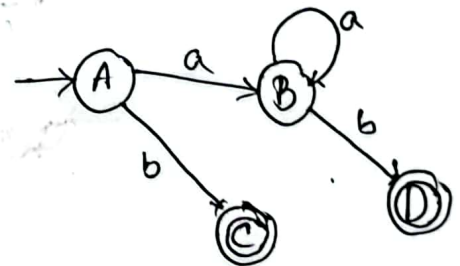
Move ($\{D\}, a$) $\rightarrow \emptyset$

Move ($\{D\}, b$) $\rightarrow \emptyset$

ϵ -closure ($\{A, 7\}$) $\rightarrow \{4, 7\} \rightarrow C$

Move ($\{C\}, a$) $\rightarrow \emptyset$

Move ($\{C\}, b$) $\rightarrow \emptyset$



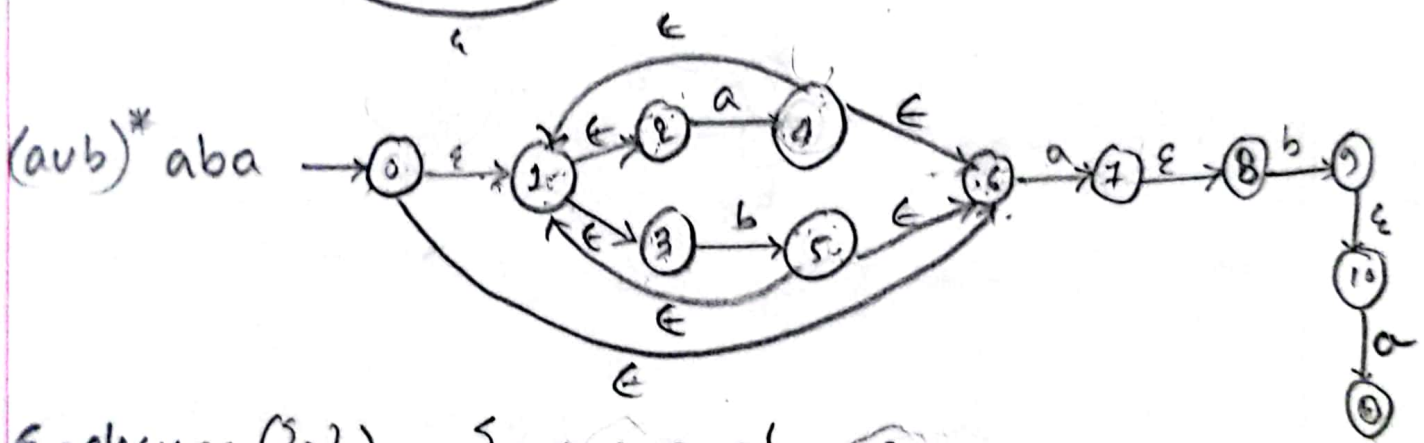
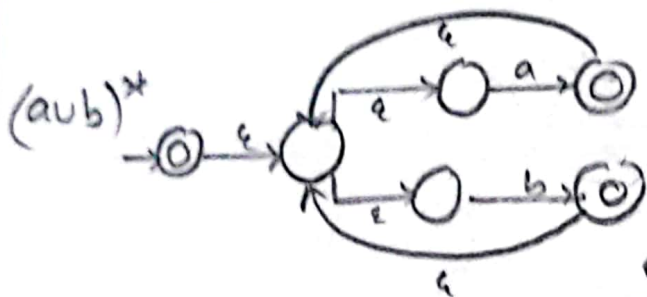
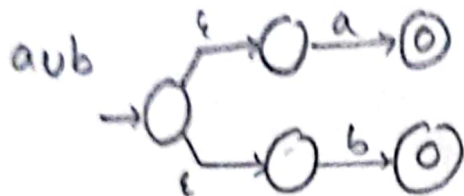
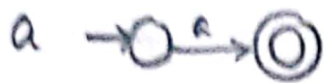
Quiz - 1

Next week

Tuesday Sunday

RE $\rightarrow$ NFA $\rightarrow$ DFA

$(a \cup b)^* aba$



ϵ -closure($\{0\}$) $\rightarrow \{0, 1, 2, 3, 6\} \rightarrow A$

move($\{A\}, a$) $\rightarrow \{4, 7\} \rightarrow B$ ✓

move($\{A\}, b$) $\rightarrow \{5\} \rightarrow C$

ϵ -closure($\{4, 7\}$) $\rightarrow \{1, 2, 3, 4, 6, 7, 8\} \rightarrow B$

move($\{B\}, a$) $\rightarrow \{4, 7\} \rightarrow B$

move($\{B\}, b$) $\rightarrow \{5, 9\} \rightarrow D$

ϵ -closure($\{5\}$) $\rightarrow \{1, 2, 3, 5, 6\} \rightarrow C$

move($\{C\}, a$) $\rightarrow \{4, 7\} \rightarrow B$

move($\{C\}, b$) $\rightarrow \{5\} \rightarrow C$

$\epsilon\text{-closure}(\{5, 0\}) \rightarrow \{1, 2, 3, 5, 6, 9, 10\} \rightarrow D$

$\text{move}(\{D\}, a) \rightarrow \{4, 7, 11\} \rightarrow E$

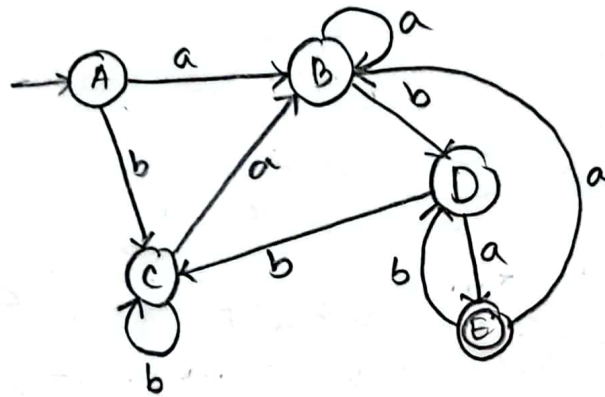
$\text{move}(\{D\}, b) \rightarrow \{5\} \rightarrow C$

$\epsilon\text{-closure}(\{4, 7, 11\}) \rightarrow \{1, 2, 3, 4, 6, 7, 8, 11\} \rightarrow E$

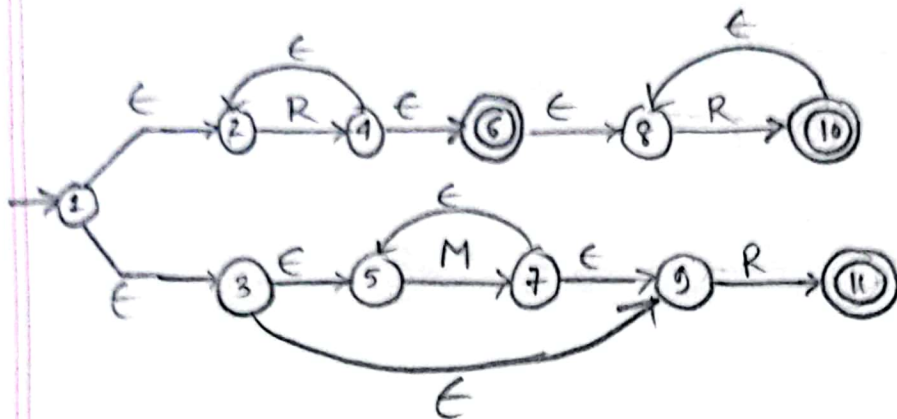
$\text{move}(\{E\}, a) \rightarrow \{1, 7\} \rightarrow B$

$\text{move}(\{E\}, b) \rightarrow \{5, 0\} \rightarrow D$

DFA



$R^+R^* | M^*R$



ϵ -closure ($\{1\}$) $\rightarrow \{1, 2, 3, 5, 9\} \rightarrow A$

move ($\{A\}, R$) $\rightarrow \{4, 11\} \rightarrow B$

move ($\{A\}, M$) $\rightarrow \{7\} \rightarrow C$

ϵ -closure ($\{4, 11\}$) $\rightarrow \{2, 4, 6, 8, 11\} \rightarrow B$

move ($\{B\}, R$) $\rightarrow \{4, 10\} \rightarrow D$

move ($\{B\}, M$) $\rightarrow \emptyset$

ϵ -closure ($\{7\}$) $\rightarrow \{5, 9\} \rightarrow C$

move ($\{C\}, R$) $\rightarrow \{11\} \rightarrow E$

move ($\{C\}, M$) $\rightarrow \{7\} \rightarrow C$

ϵ -closure ($\{4, 10\}$) $\rightarrow \{2, 4, 6, 8, 10\} \rightarrow D$

move ($\{D\}, R$) $\rightarrow \{4, 10\} \rightarrow D$

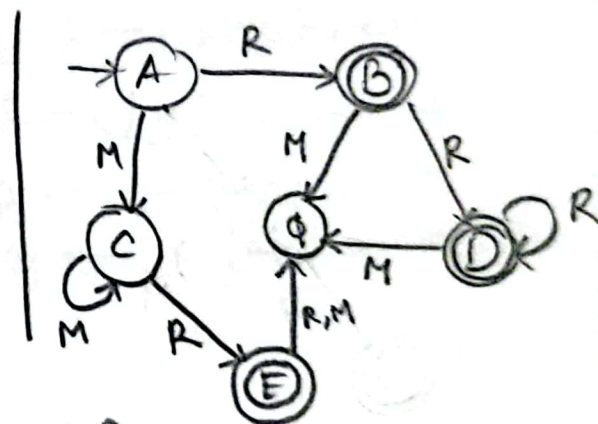
move ($\{D\}, M$) $\rightarrow \emptyset$

ϵ -closure ($\{11\}$) $\rightarrow \{11\} \rightarrow E$

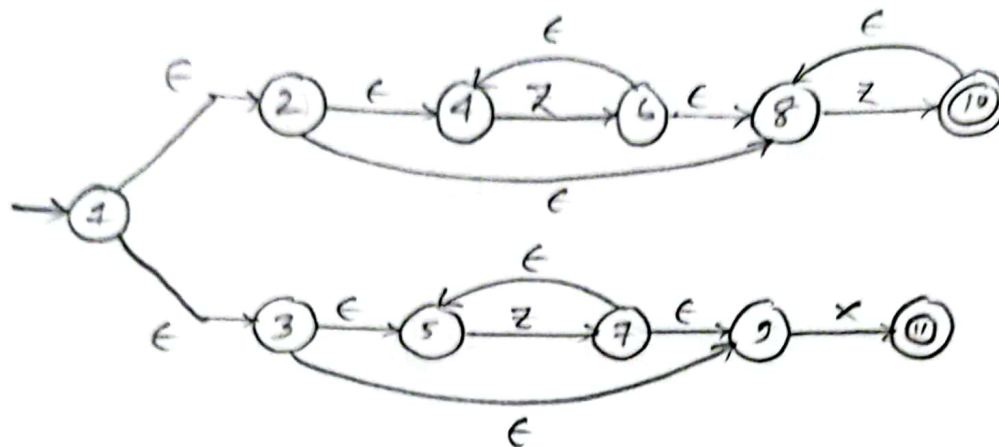
move ($\{E\}, R$) $\rightarrow \emptyset$

move ($\{E\}, M$) $\rightarrow \emptyset$

DFA



$z^* z^+ | z^* x$



$\epsilon\text{-closure}(\{1\}) \rightarrow \{1, 2, 3, 4, 5, 8, 9\} - A$

$\text{move}(\{A\}, z) \rightarrow \{6, 7, 10\} \rightarrow B$

$\text{move}(\{A\}, x) \rightarrow \{11\} - C$

$\epsilon\text{-closure}(\{6, 7, 10\}) \rightarrow \{1, 5, 6, 7, 8, 9, 10\} - B$

$\text{move}(\{B\}, z) \rightarrow \{6, 7, 10\} \rightarrow B$

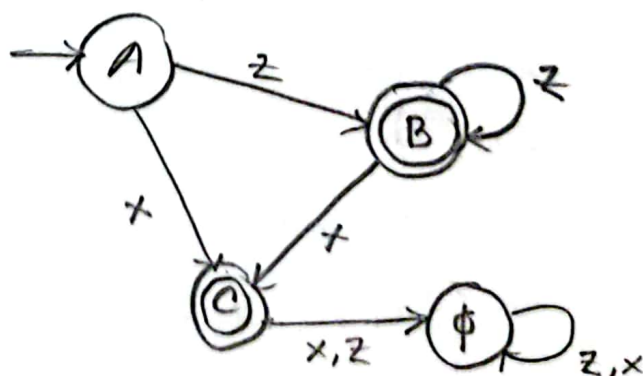
$\text{move}(\{B\}, x) \rightarrow \{11\} - C$

$\epsilon\text{-closure}(\{11\}) \rightarrow \{11\} - C$

$\text{move}(\{C\}, z) \rightarrow \phi$

$\text{move}(\{C\}, x) \rightarrow \phi$

DFA



First & Follow

must be from CFG

First

যদি first element এর production rule এর first element হয় তবে variable এর First

 $E \rightarrow @T$ First $(E) = \{ @ \}$ → যদি terminal হয় তাহলে direct বসে যাবে। $E \rightarrow @T / \epsilon$ First $(E) = \{ @, \epsilon \}$

When the first element is non-terminal.

 $E \rightarrow FT$ First $(E) = \text{First}(F)$ $F = \{ a, b \}$

Exception:

 $F = \{ a, b, \epsilon \}$ & $T = \{ c \}$ $E \rightarrow FT = \text{First}(T)$ First $(E) = \{ c \}$

P. rule

First

Follow

 $E \rightarrow TE'$ $\{ c, id \}$ $E' \rightarrow +TE' / \epsilon$ $\{ +, \epsilon \}$ $T \rightarrow FT'$ $\{ c, id \}$ $T' \rightarrow *FT' / \epsilon$ $\{ *, \epsilon \}$ $F \rightarrow (E) / id$ $\{ c, id \}$ $S \rightarrow ACB$ $\{ c, b, a \}$ $A \rightarrow da / BC$ $\{ d, \epsilon \}$ $B \rightarrow g / \epsilon$ $\{ g, \epsilon \}$ $C \rightarrow h / \epsilon$ $\{ h, \epsilon \}$ Follow $(E) = \{ \$,) \}$ Follow $(E') = \{ \$,) \}$ Follow $(T) = \{ +, \$,) \}$ Follow $(T') = \{ +, \$,) \}$ Follow $(F) = \{ (, \$,) \}$ Follow $(S) = \{ \$ \}$ Follow $(A) = \{ h, g, \$ \}$ Follow $(B) = \{ \$, a, h, g \}$ Follow $(C) = \{ g, \$, b, h \}$

যদি follow বসে থাকে
হবে

Follow → Production rule এর কোন মুহুর্তে হবে,
কখনো ϵ হবে না।

Case 1 $T \rightarrow E a$ → start variable এর জন্য follow set এর মধ্যে
by default $\$$ হবে। must থাকবে
 $\text{Follow}(E) = \{a\}$ → Follow terminal

Case 2 $T \rightarrow EF$ $F = \{a, b\}$ → First of F

$\text{Follow}(E) = \{a, b\}$ → when follow is a non-terminal,
হবে তাহলে, এর non-terminal এর
first কখনো দেখতে হবে।

Case 3 $T \rightarrow E$ → যদি কোনো follow না থাকে তাহলে তার
Parent এর follow হ'বে তার follow
 $\text{Follow}(E) = \text{Follow}(T)$

$T \rightarrow EF$ $\text{Follow}(F) = \{a, \epsilon\}$

$T \rightarrow E \cancel{F}$ $\text{Follow}(T) = \{b\}$

$\text{Follow}(E) = \{a, b\}$

	First	Follow
$S \rightarrow XY W$	$\{ @, \%, \epsilon, \$ \}$	$\{ \$ \}$
$X \rightarrow @X\% \epsilon$	$\{ @, \epsilon \}$	$\{ \%, \$ \}$
$Y \rightarrow \%Y \epsilon$	$\{ \%, \epsilon \}$	$\{ \$ \}$
$W \rightarrow @W\% z$	$\{ @, \$, \epsilon \}$	$\{ \$, \% \}$
$z \rightarrow \$z \epsilon$	$\{ \$, \epsilon \}$	$\{ \$, \% \}$